Escuela Superior Politécnica del Litoral (ESPOL)
Guayaquil, Ecuador

October 11, 2026

The Editors
Scientific Reports

Dear Editors,

We submit the manuscript "Few biological parameters suffice to reproduce measured dynamics in systems biology models: local sensitivity-based reduction with formally verified mathematical guarantees" by Mónica Miranda R., Carlos Salazar, César Martín and Adriana Aguirre for consideration as an Article in *Scientific Reports*.

Mathematical models of biological and behavioural processes often contain far more unknown parameters than experimental data can determine. We present a method that, before any model fitting, selects a small set of model parameters that is sufficient to reproduce the measured data, with all other parameters kept at their published values. Applied to 34 published models with real experimental data from the PEtab benchmark collection, the method found such a subset in 32 models, in most cases only two parameters. The reduced models predicted held-out experiments as well as or better than the full models in 27 of 28 testable cases, and 27 of the 32 reductions remained valid for parameter changes of up to 50%. The main mathematical steps of the method are verified with the Lean proof assistant, and all code and results are openly available.

We believe the work suits *Scientific Reports* because it addresses a problem shared by many fields that use dynamical models, including systems biology, pharmacology, epidemiology and behavioural science, and because its conclusions rest on a large, public benchmark and on reproducible code rather than on a single case study. The study was motivated by a social cognitive theory model of physical activity, which is included as an illustrative case.

Suggested reviewers. [Name, affiliation, e-mail and area of expertise of three to five suggested reviewers.]

Excluded reviewers. [Names of any referees to be excluded, or "None".]

Prior discussions. We have had no prior discussions with any Editorial Board Member of *Scientific Reports* about this work.

Related submissions. [State here the status of any related conference or journal submission of this work, e.g. withdrawn or not overlapping.]

All authors have approved the manuscript and agree with its submission. The corresponding author

is Mónica Miranda R.

Sincerely,

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